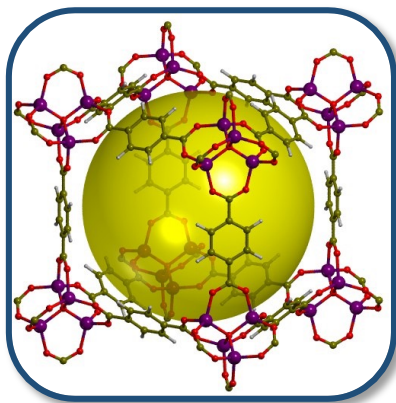
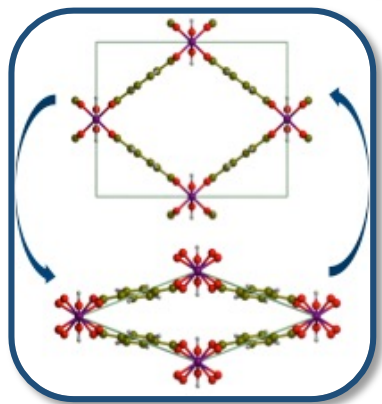
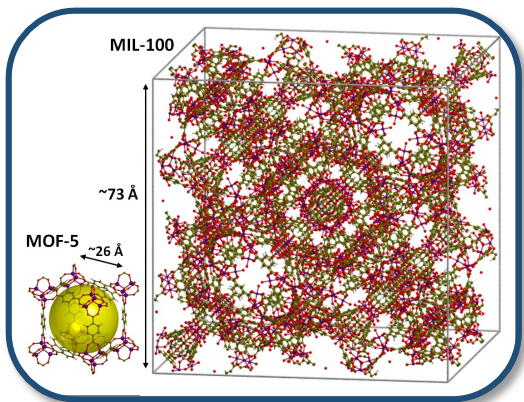


MSSC2025 - *Ab Initio* Modelling in Solid State Chemistry

TORINO 8-12 SEP 2025

How to Model Metal-Organic Frameworks with CRYSTAL



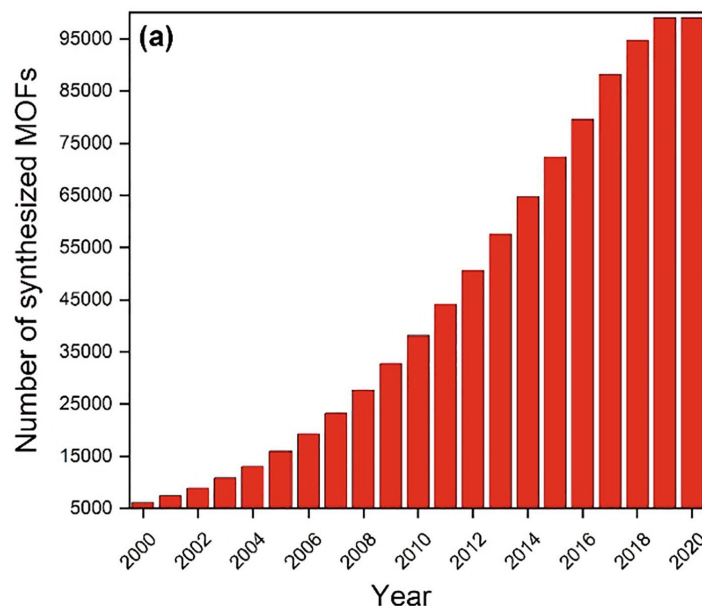
Donà Lorenzo

Department of Chemistry
University of Torino,
Italy



A rapidly growing class of versatile hybrid organic-inorganic materials

**MOFs are relatively new materials (O. Yaghi 1995-1999)
In the last decades, they have seen a paramount growth
(nr. of MOFs structure in CSD – **98218 in 2024**)**

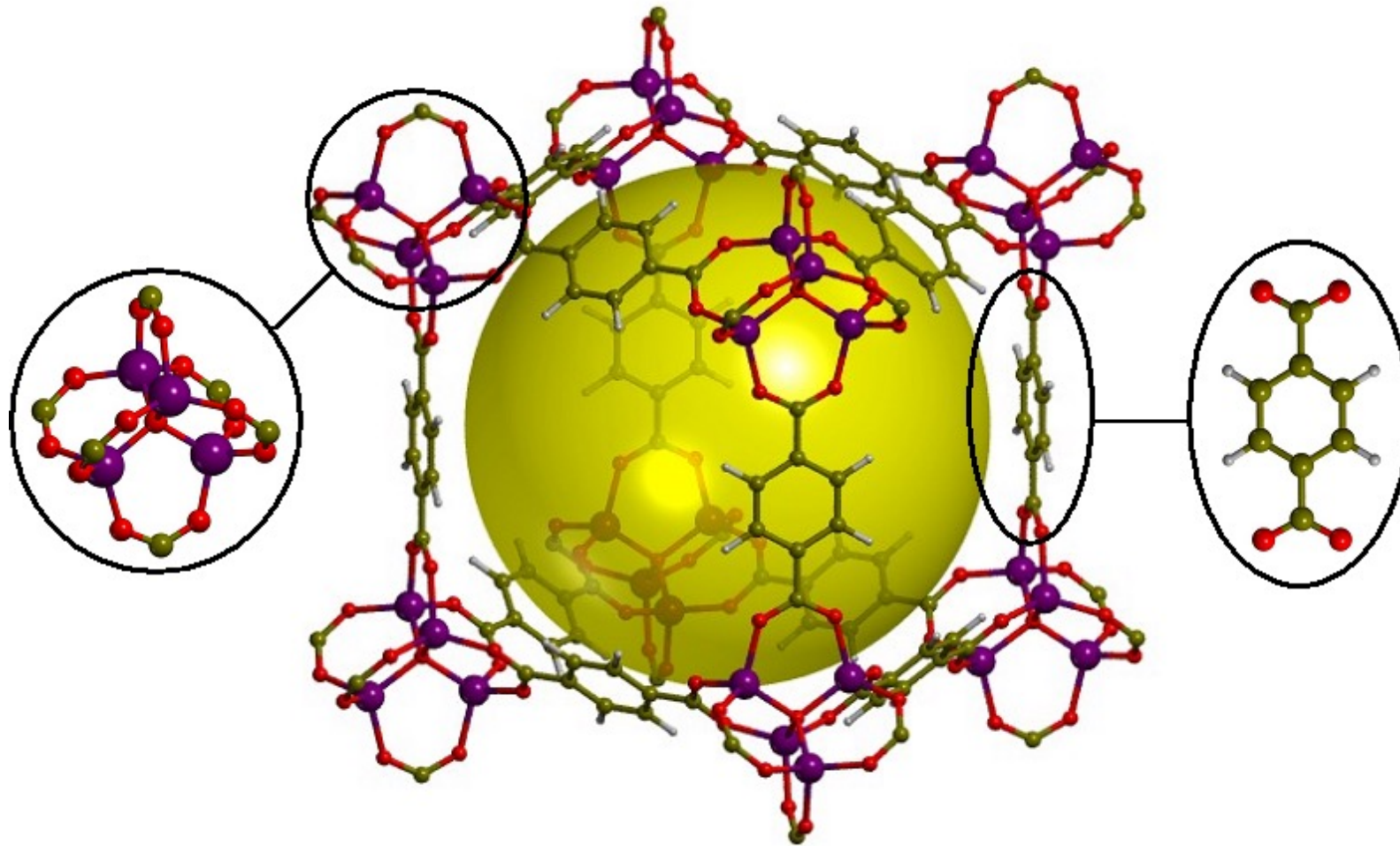


Recent advances, opportunities, and challenges in high-throughput computational screening of MOFs for gas separations.
from H. Daglar, S. Keskin, Coordination Chemistry Reviews 422 (2020) 213470

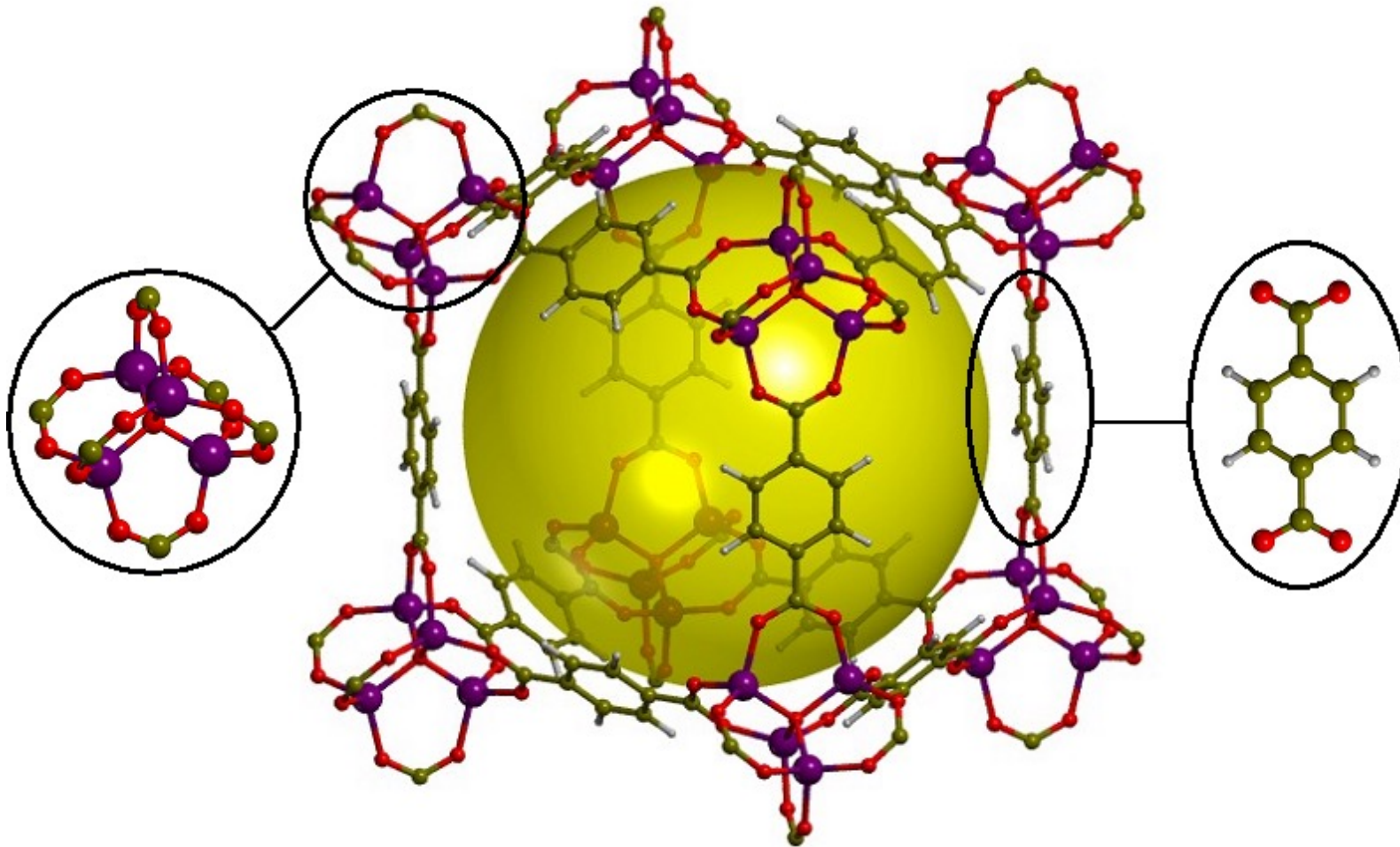
Huge impact in the development of next-generation technologies



Metal-Organic Frameworks (MOFs)



Metal-Organic Frameworks (MOFs)



- Unique combination of structure and bonding between inorganic part (e.g. *metal*) and organic ligands (e.g. *carboxylate*)

- 3D crystalline structure *stable, rigid and porous* by **rational design**
- **Chemical versatility** (Tinkertoy materials)

Levels of complexity in MOFs



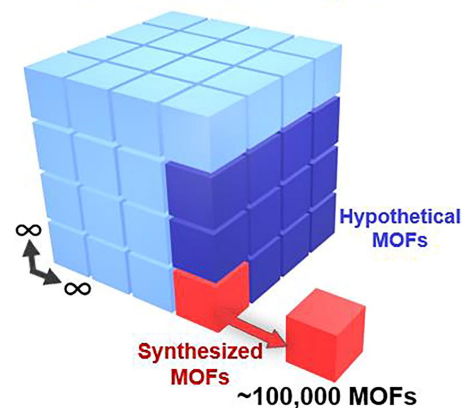
***Three intrinsic
degrees of
complexity in
MOFs***

Levels of complexity in MOFs



The exceptional *chemical versatility* leads to **a plethora of different structures**: from both synthesis and material design (HTCS, CSP)

Synthesized MOF ≈ 100000
MOF Material Space

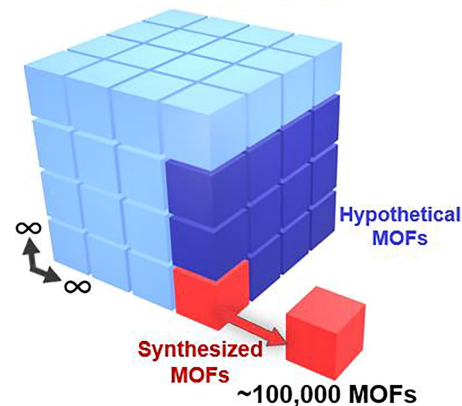


Three intrinsic degrees of complexity in MOFs

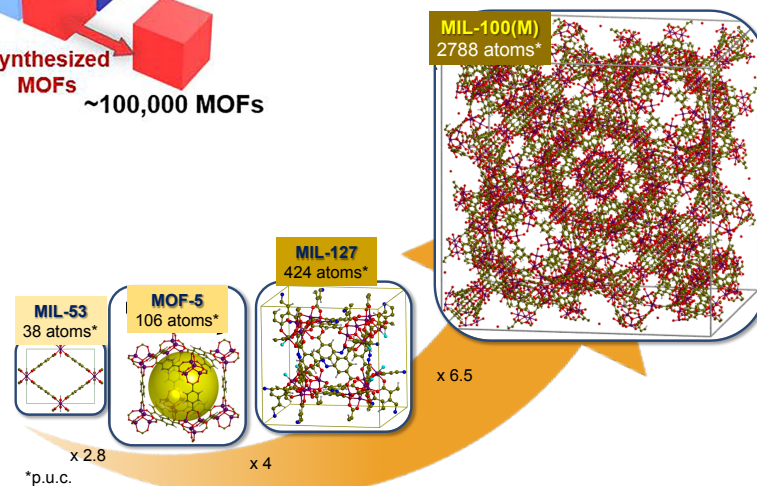
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The **size of the system** can pose a limit to simulation: from 20-30 to 10^3 atoms in the unit cell

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MOF Material Space

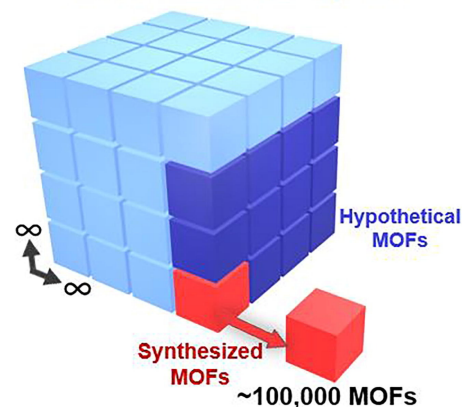


Three intrinsic degrees of complexity in MOFs

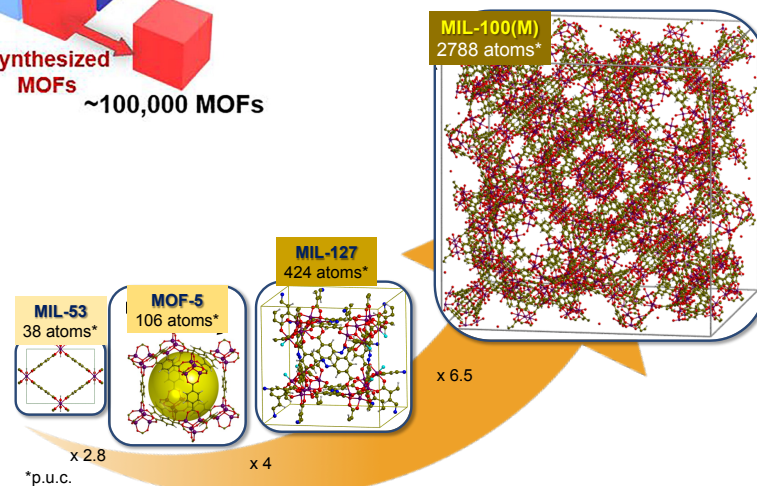


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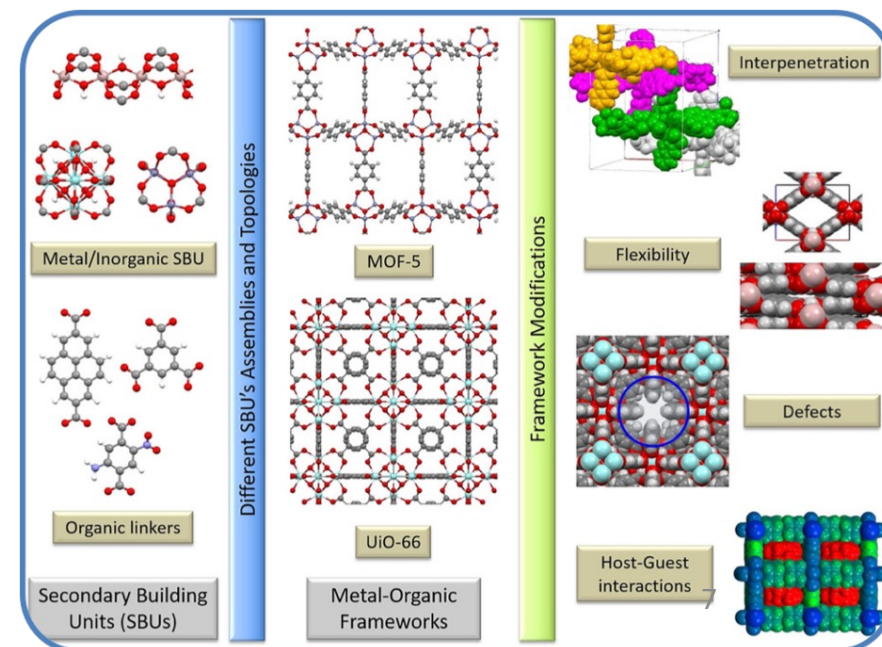


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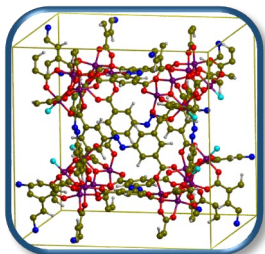


Structural complexity: from secondary building units to quaternary supramolecular structures; from defects to host-guest interactions

Three intrinsic degrees of complexity in MOFs



MOFs as a playground for CRYSTAL

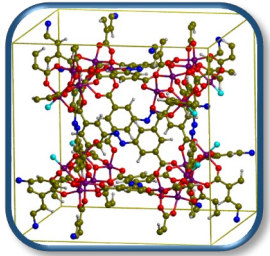


MOFs

CRYSTAL possesses various features and tools that makes it very effective for investigating MOFs



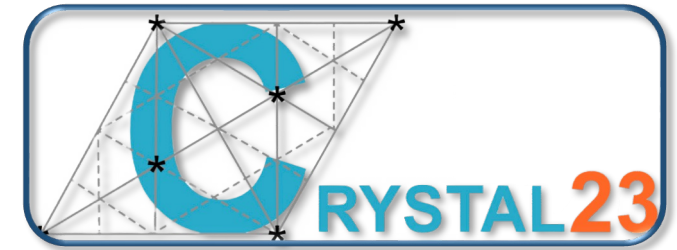
CRYSTAL



MOFs

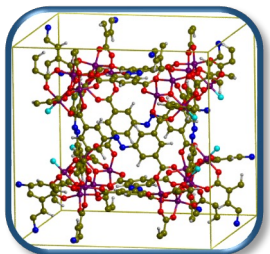
- 3D crystalline structures

CRYSTAL possesses various features and tools that makes it very effective for investigating MOFs



CRYSTAL

- 3D Periodic Boundary Conditions, k-point sampling, ...



MOFs

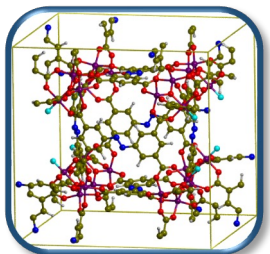
- 3D crystalline structures
- Symmetric framework

CRYSTAL possesses various features and tools that makes it very effective for investigating MOFs



CRYSTAL

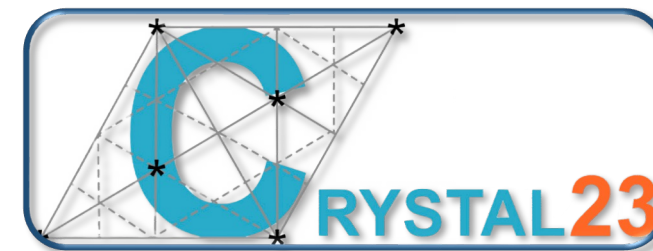
- 3D Periodic Boundary Conditions, k-point sampling, ...
- Full exploitation of the symmetry



MOFs

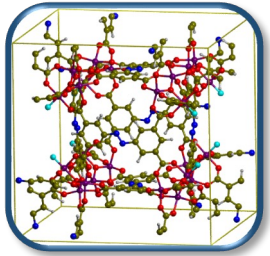
- 3D crystalline structures
- Symmetric framework
- Porous materials (voids)

CRYSTAL possesses various features and tools that makes it very effective for investigating MOFs



CRYSTAL

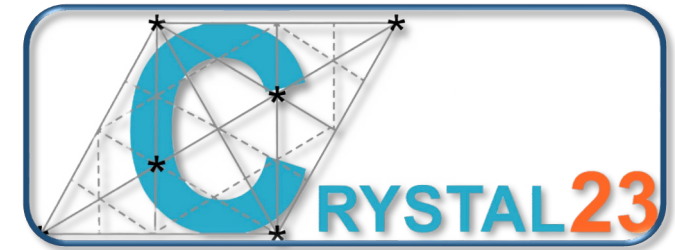
- 3D Periodic Boundary Conditions, k-point sampling, ...
- Full exploitation of the symmetry
- Atom-centered basis functions vs PWs



MOFs

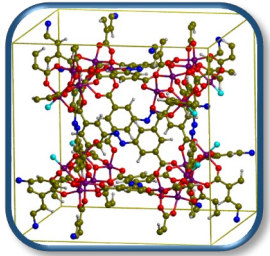
- 3D crystalline structures
- Symmetric framework
- Porous materials (voids)
- Large unit cells or low symmetry

CRYSTAL possesses various features and tools that makes it very effective for investigating MOFs



CRYSTAL

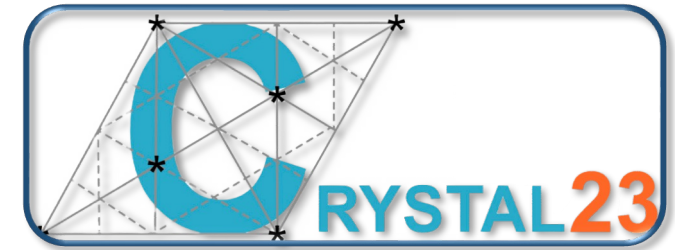
- 3D Periodic Boundary Conditions, k-point sampling, ...
- Full exploitation of the symmetry
- Atom-centered basis functions vs PWs
- Pcrystal & MPPcrystal



MOFs

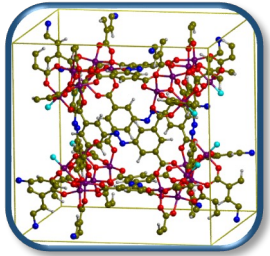
- 3D crystalline structures
- Symmetric framework
- Porous materials (voids)
- Large unit cells or low symmetry
- Adsorption and catalysis

CRYSTAL possesses various features and tools that makes it very effective for investigating MOFs



CRYSTAL

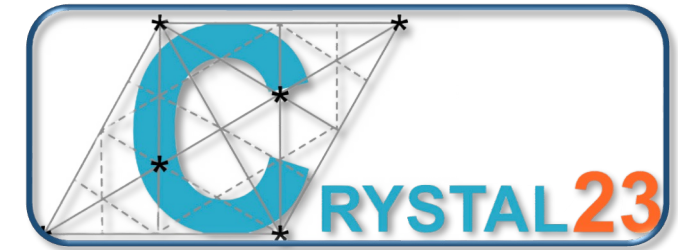
- 3D Periodic Boundary Conditions, k-point sampling, ...
- Full exploitation of the symmetry
- Atom-centered basis functions vs PWs
- Pcrystal & MPPcrystal
- Dispersion corrections on top of DFT



MOFs

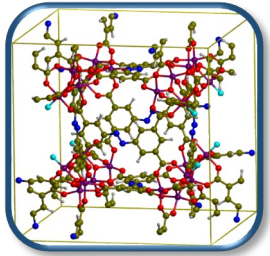
- 3D crystalline structures
- Symmetric framework
- Porous materials (voids)
- Large unit cells or low symmetry
- Adsorption and catalysis
- Mechanical & thermal stability

CRYSTAL possesses various features and tools that makes it very effective for investigating MOFs



CRYSTAL

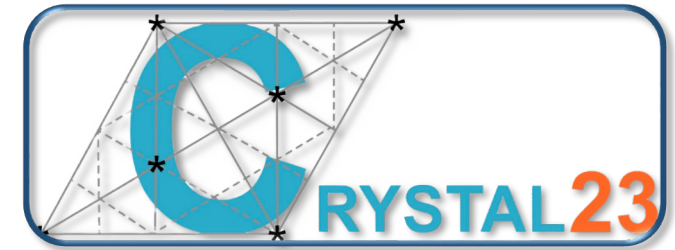
- 3D Periodic Boundary Conditions, k-point sampling, ...
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- Dispersion corrections on top of DFT
- Automated tools: EOS, ELASTCON, QHA



MOFs

- 3D crystalline structures
- Symmetric framework
- Porous materials (voids)
- Large unit cells or low symmetry
- Adsorption and catalysis
- Mechanical & thermal stability
- Electronic properties and derived functions

CRYSTAL possesses various features and tools that makes it very effective for investigating MOFs



CRYSTAL

- 3D Periodic Boundary Conditions, k-point sampling, ...
- Full exploitation of the symmetry
- Atom-centered basis functions vs PWs
- Pcrystal & MPPcrystal
- Dispersion corrections on top of DFT
- Automated tools: EOS, ELASTCON, QHA
- Hybrid functionals (band gaps, CPHF₁₅...)

Hints to model MOFs with CRYSTAL



- **Crystalline structure**

- from *cif* files
- Databases are available (e.g. CoRE-MOF 2019 many structures without symmetry; QMOF 2021 tested through QM calc.s)

- Carefully check the geometry: atomic positions (e.g. H), partial occupancy, counter ions, water molecules, solvent
- Sometimes the experimental structure is too symmetric therefore a symmetry reduction is needed to remove structural disorder

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- **Methods**

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Hints to model MOFs with CRYSTAL



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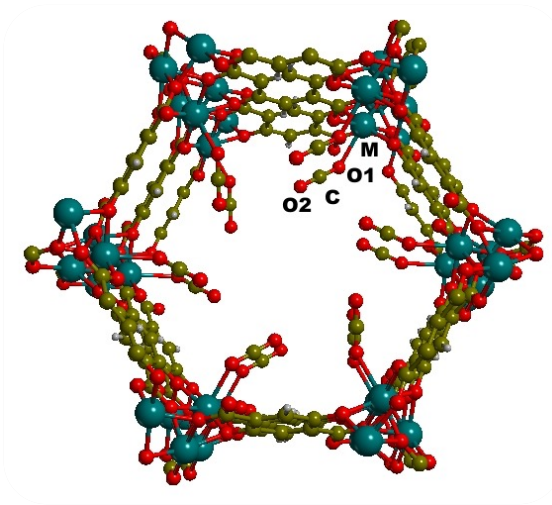
- **Methods**

- **Basis sets:**

- Organic linker
- Metal ion

- Carefully check the geometry: atomic positions (e.g. H), partial occupancy, counter ions, water molecules, solvent
- Sometimes the experimental structure is too symmetric therefore a symmetry reduction is needed to remove structural disorder
- Hybrid functionals including dispersion corrections (D2 or D3)
- Usually, molecular basis sets work fine (e.g. 6-311G(d,p))
- Basis set must be adapted to the electronic configuration of the metal (reoptimization or pruning)
- POB basis sets (DZ or TZ quality for most of the elements)

Metal

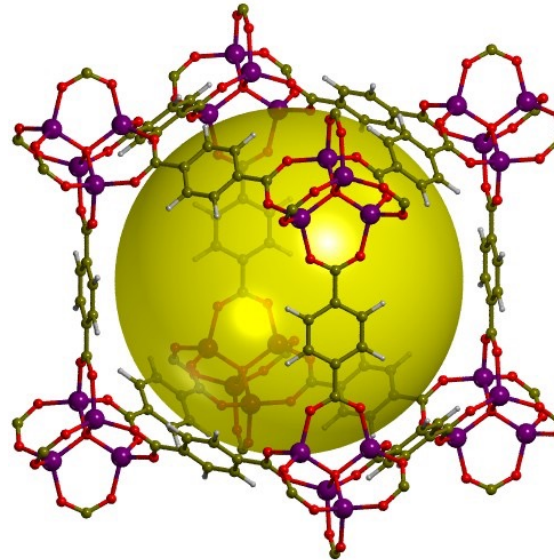


Role of the metal

Adsorption
properties

Open metal sites
MIL-101 (Cr)

Organic

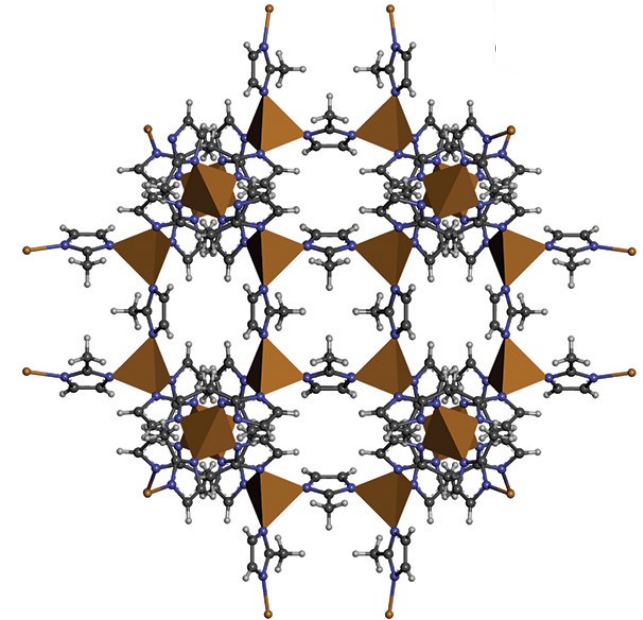


Role of the linker

Electronic/dielectric
properties

Functionalization
IRMOF-1 family

Framework

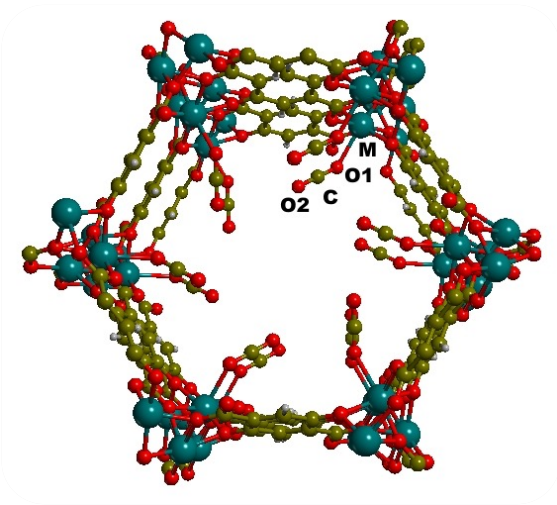


Interplay between organic & inorganic part

Elastic & Vibrational
properties

Defect engineering
ZIF-8 (Zn)

Metal



Role of the metal

Adsorption
properties

Open metal sites
MIL-101 (Cr)

How far can we go with MOFs?



Low-cost composite methods

PBEsol0-3c

(hybrid HF/DFT: 25%)

**Accuracy comparable to
TZ quality calculations**

$$E_{tot}^{(3c)sol3c} = E_{tot}^{HF,DFT/basis} + E_{disp}^{D3} + E_{BSSE}^{gCP} + E_{SRB}$$

XC functionals for solids (e.g. PBEsol0, HSEsol)

*Revised basis set: **sol-Def2-mSVP***

Refitted parameters for D3 and gCP

L. Donà, J.G. Brandenburg, and B. Civalleri, *J. Chem. Phys.* **151** (2019) 121101



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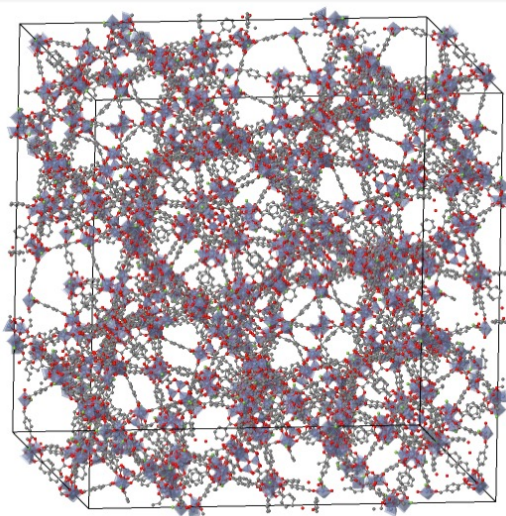
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MIL-101(Cr)/CO₂



MIL-101 (Cr)

ATOMS = 3604

AO = 46716

$a(\text{exp.})$
88.869 Å

MIL101 Lattice

Parameter (Å)

a (calc.)

88.843

a (exp.)

88.869

CO₂ Interaction

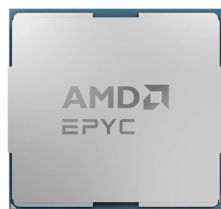
Energy (kJ/mol)

ΔE (calc.)

-40

ΔH (exp.)

-44



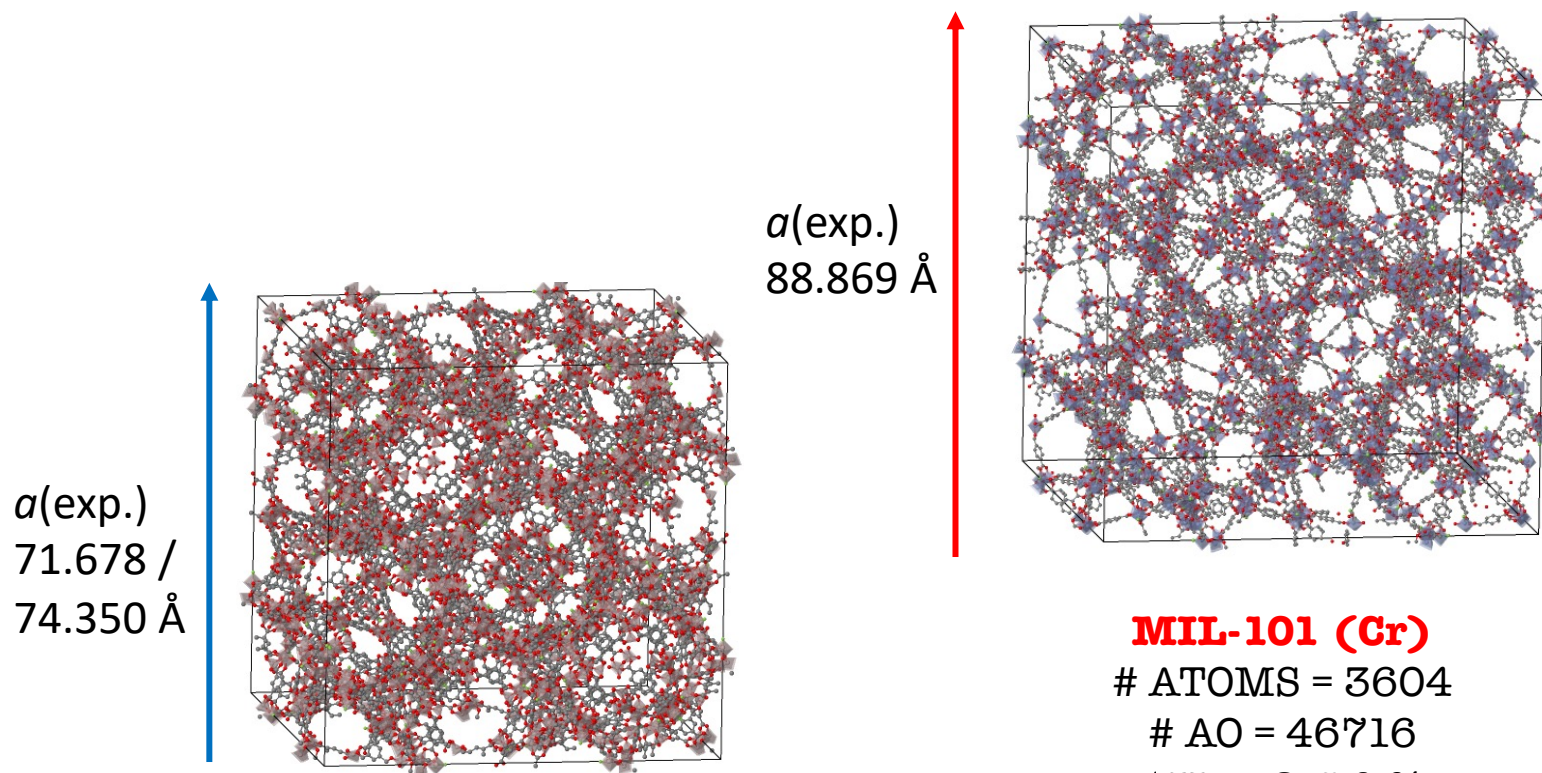
PBEsol0-3c
64 cores

Hybrid MPI/OMP
≈ 1 h SCF+GRAD

L. Donà, J.G. Brandenburg, and B. Civalleri, *JCP* **156**, 094706 (2022)



When “giant” is not enough for CRYSTAL: MIL-101-NDC



MIL-100 (M= Al, Sc, Cr, Fe)

ATOMS = 2788

AO = 39780

$\Delta V = -\mathbf{0.22} / -\mathbf{1.92} \%$

MIL-101 (Gr)

ATOMS = 3604

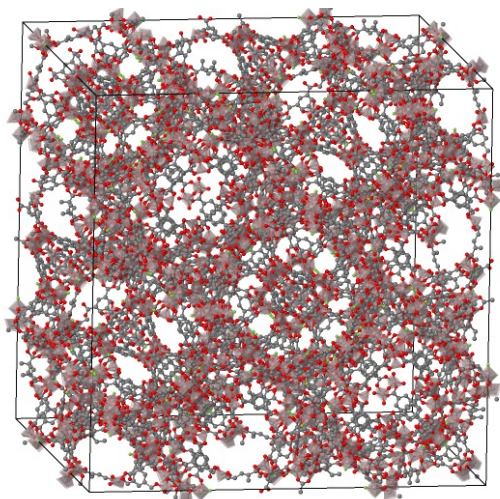
AO = 46716

$\Delta V = -\mathbf{0.14} \%$

When “giant” is not enough for CRYSTAL: MIL-101-NDC



Geometry optimization of MIL-101-NDC
affordable in two weeks on 128 CPU/cores



$a(\text{exp.})$
71.678 /
74.350 Å

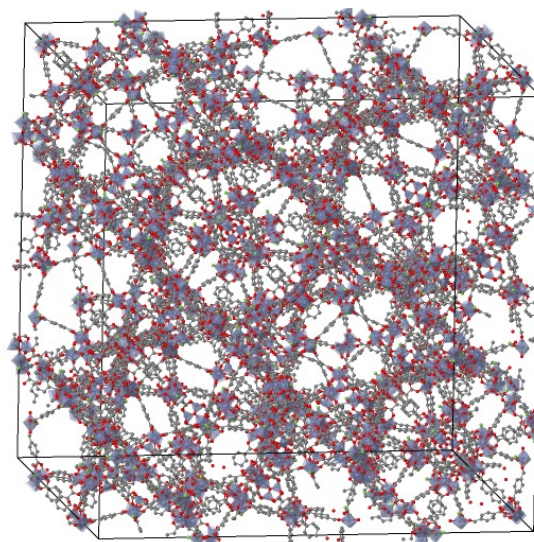
MIL-100 (M= Al, Sc, Cr, Fe)

ATOMS = 2788

AO = 39780

$\Delta V = -0.22 / -1.92 \%$

$a(\text{exp.})$
88.869 Å



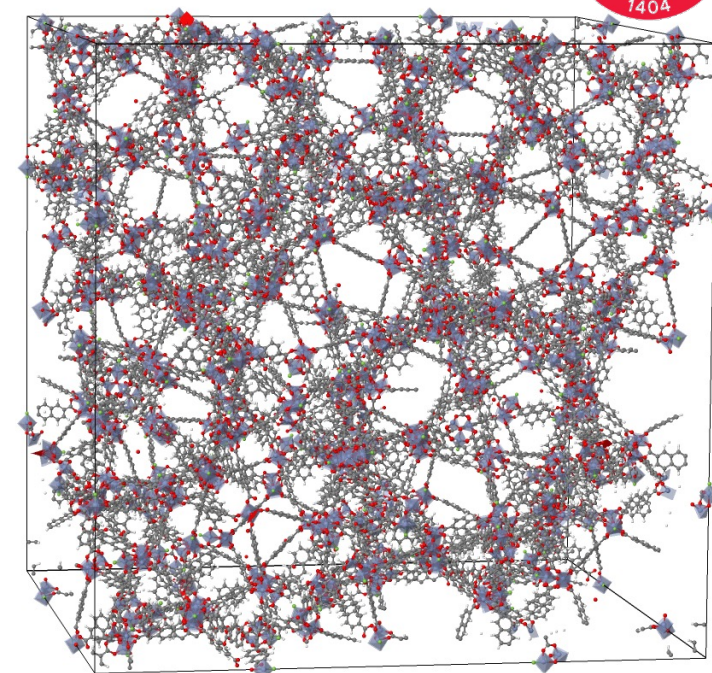
MIL-101 (Cr)

ATOMS = 3604

AO = 46716

$\Delta V = -0.14 \%$

$a(\text{exp.})$
104.5 Å



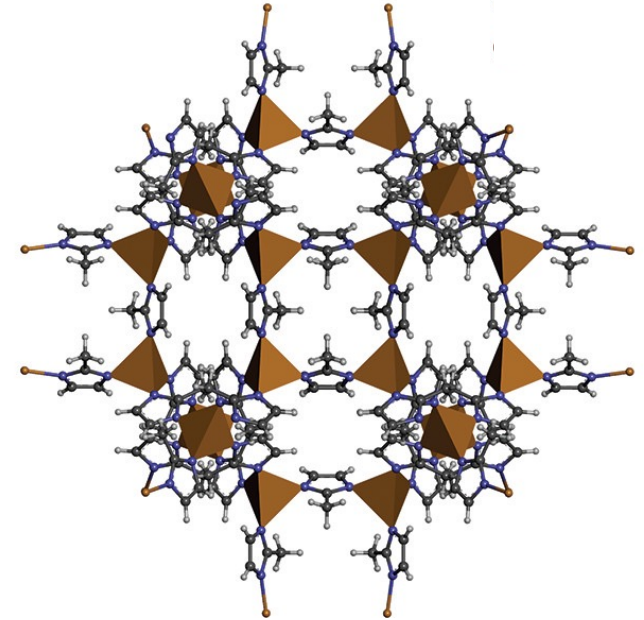
MIL-101-NDC (Cr)

ATOMS = **4828**

AO = **59772**

$\Delta V = +2.41 \%$

Framework

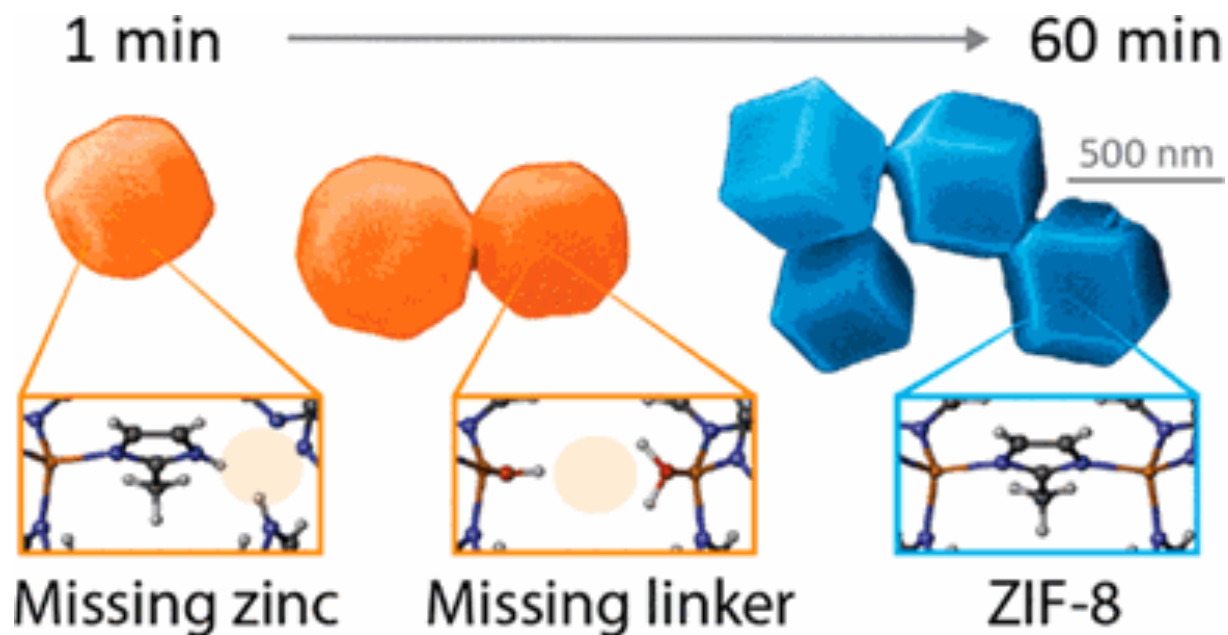
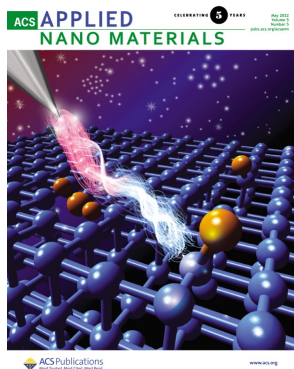


**Interplay between organic
& inorganic part**

Elastic & Vibrational
properties

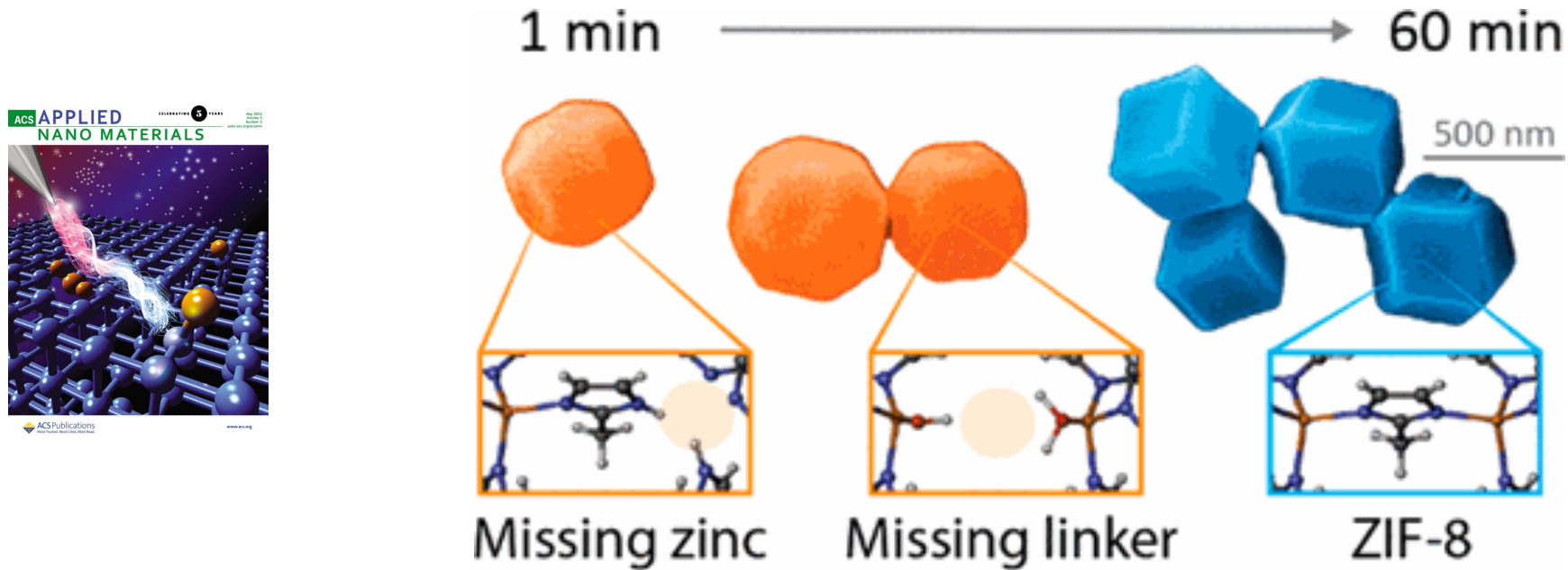
Defect engineering

ZIF-8 (Zn)



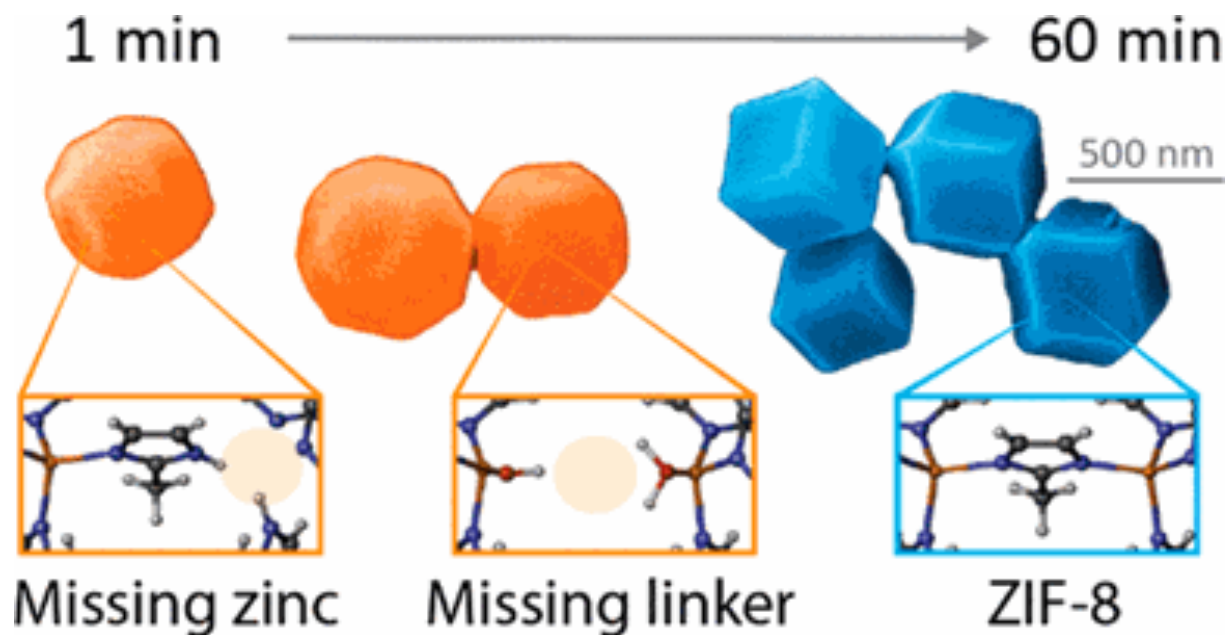
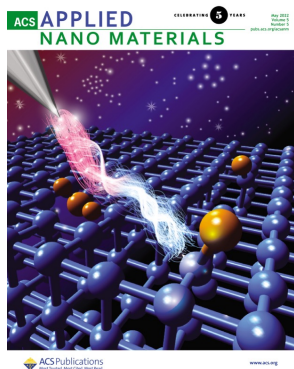
Defects in MOFs as a Challenge or an Opportunity?

Defect engineering in MOFs



Defects in MOFs as a Challenge or an Opportunity?

- Reduce the hydrophobicity, mechanical and long-term stability of the framework 😞
- Improve properties for tailored applications such as gas storage and separation, catalysis, sensing and (opto)electronics 😎



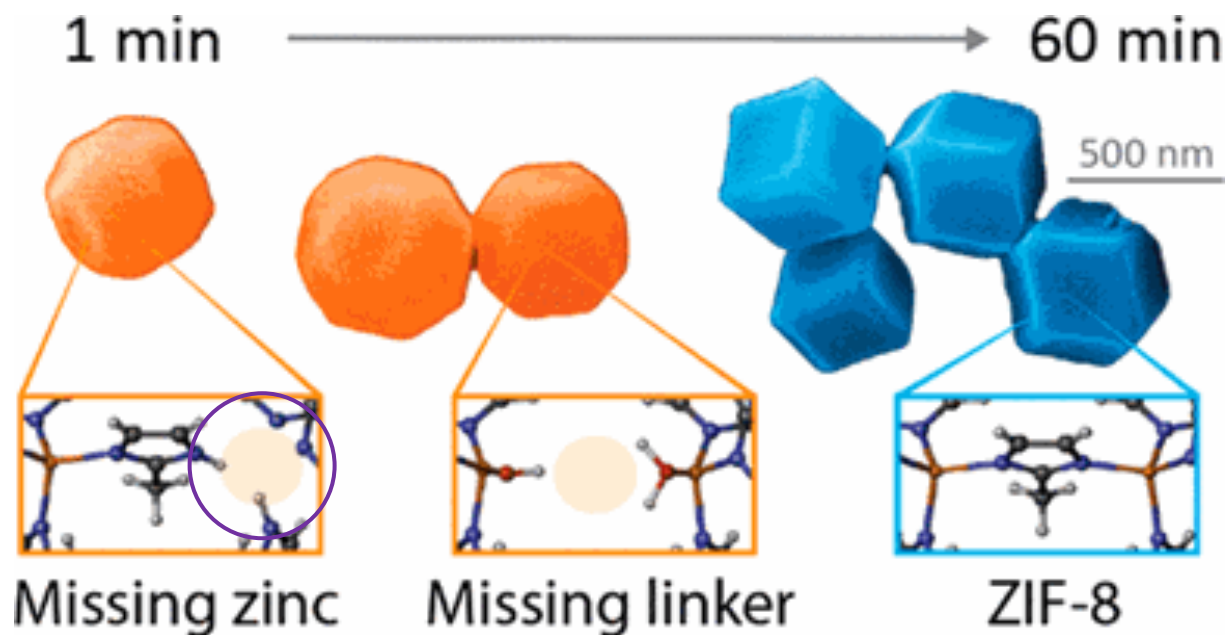
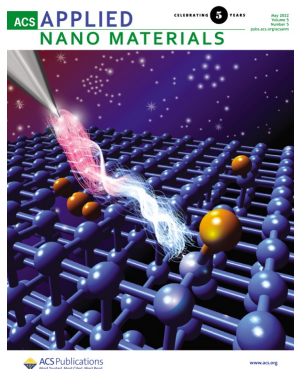
Jin-Chong Tan



Annika F. Möslin

How to model defects in ZIF-8?

- Two point defects simulated at PBEsol0-3c level of theory



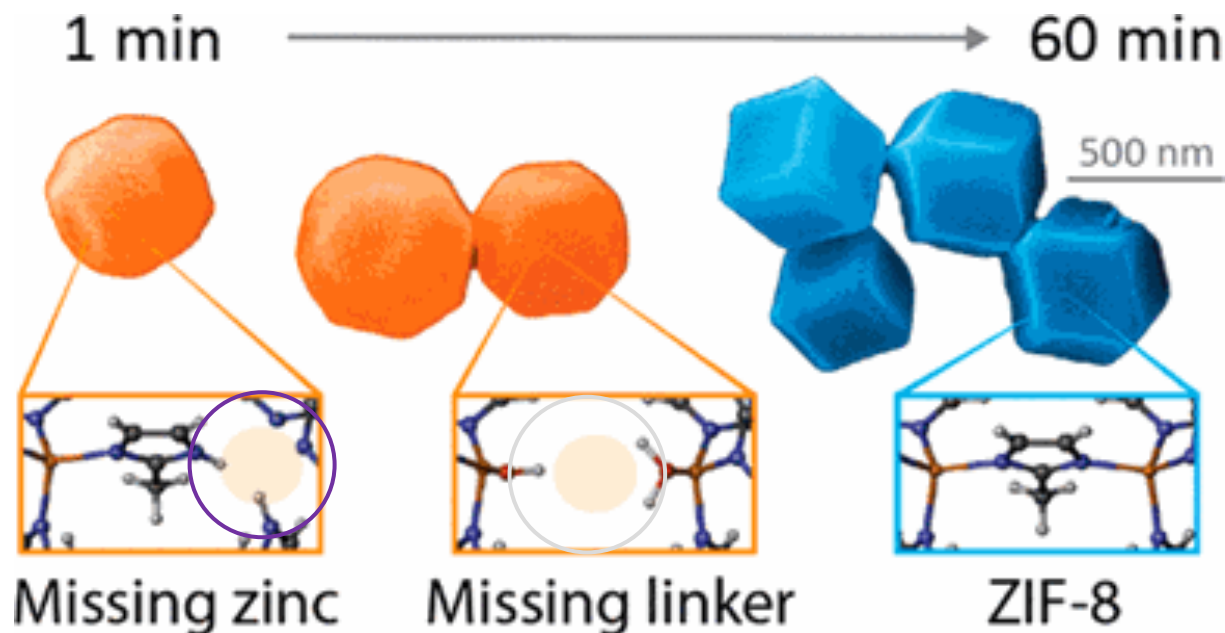
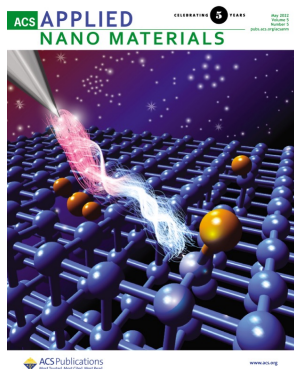
Jin-Chong Tan



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How to model defects in ZIF-8?

- Two point defects simulated at PBEsol0-3c level of theory
- Missing metal Zn-N bonds replaced by N-H bonds



Jin-Chong Tan

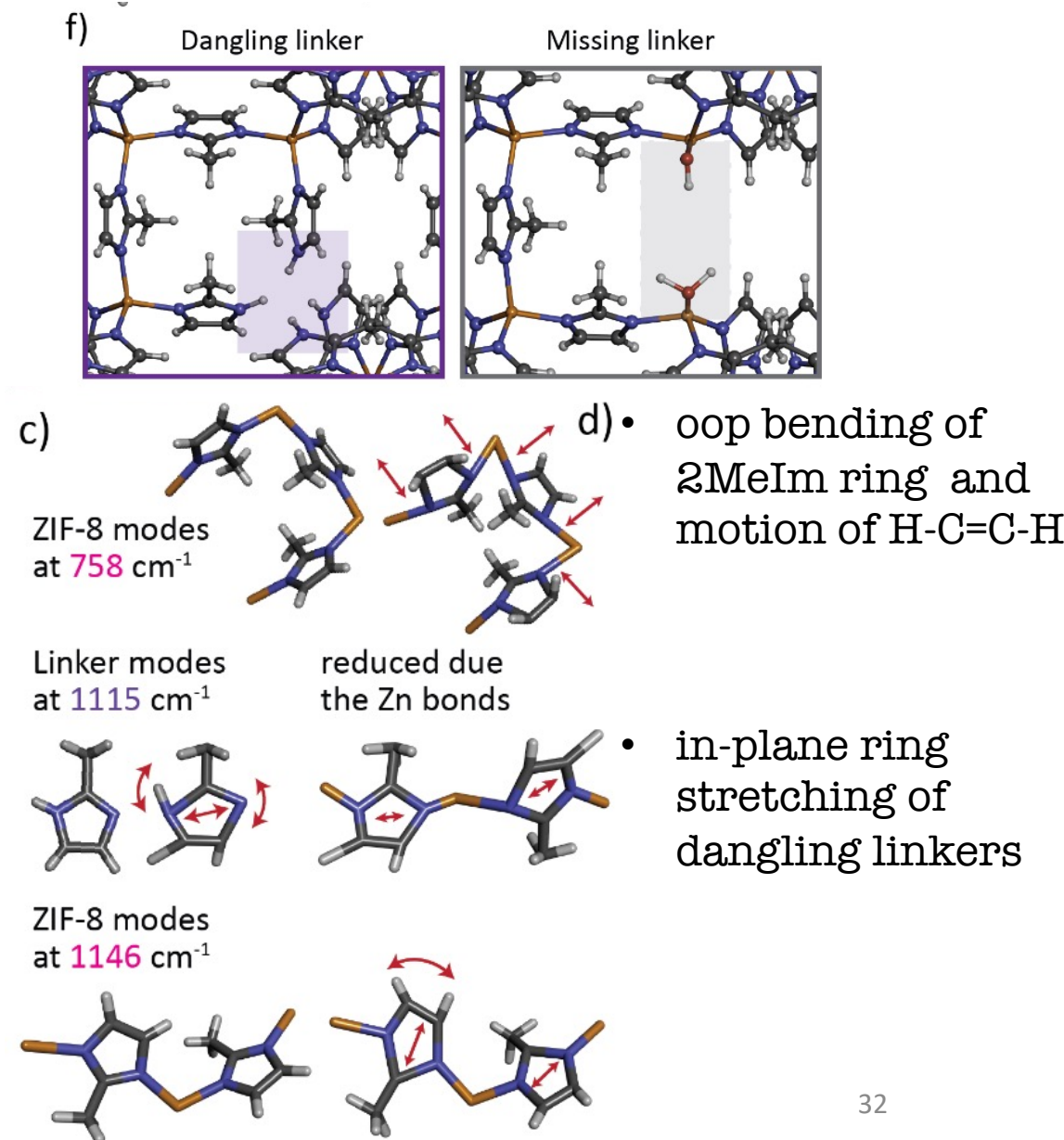
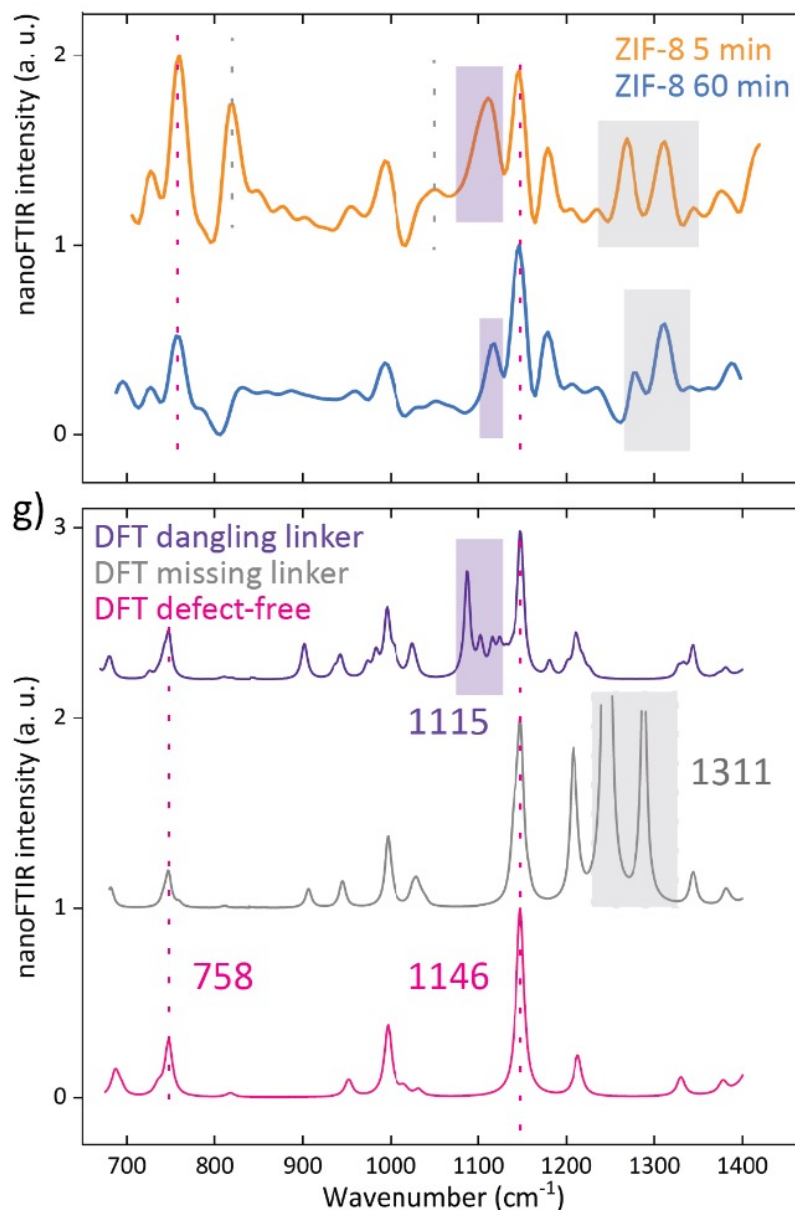


Annika F. Möslin

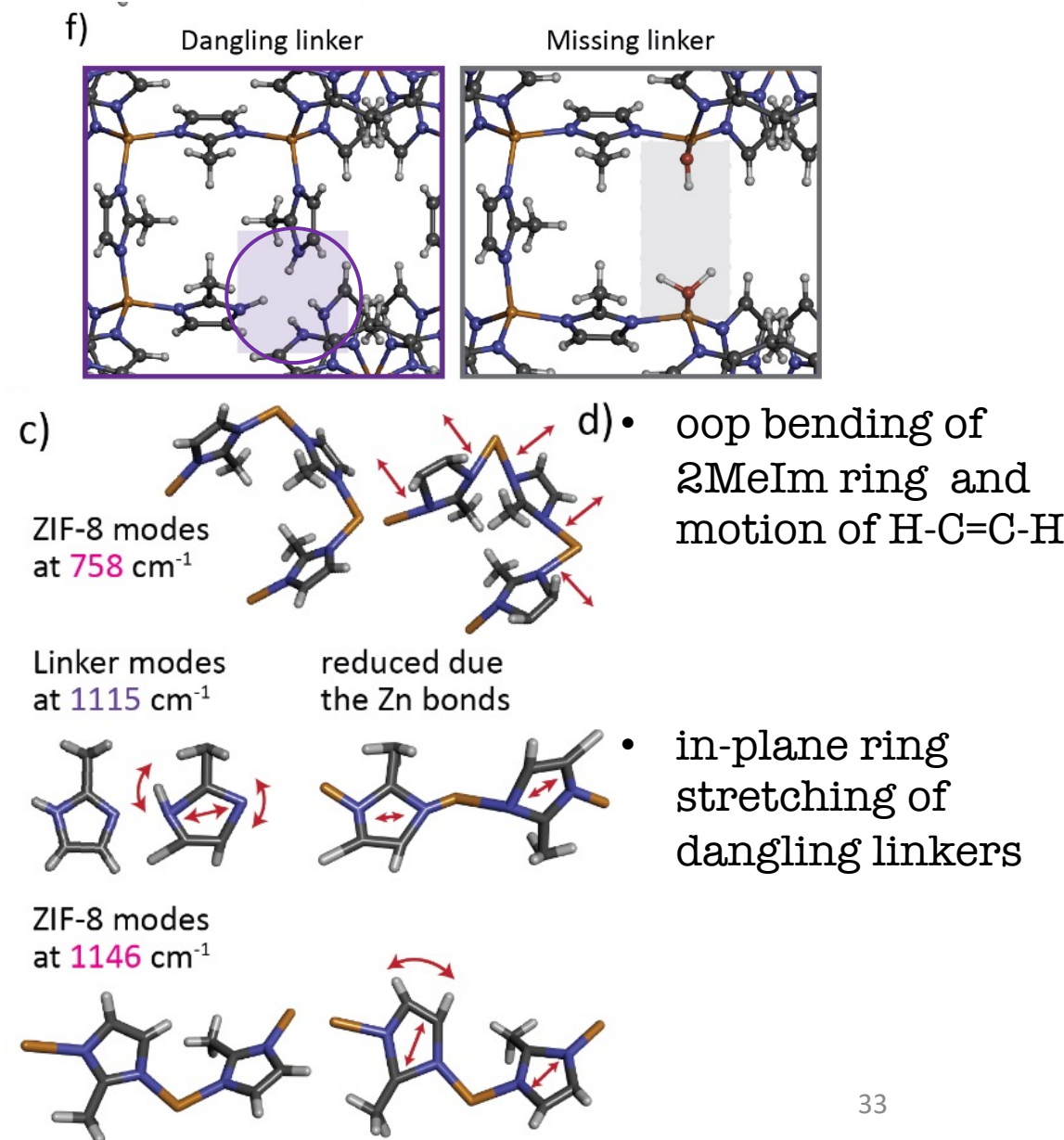
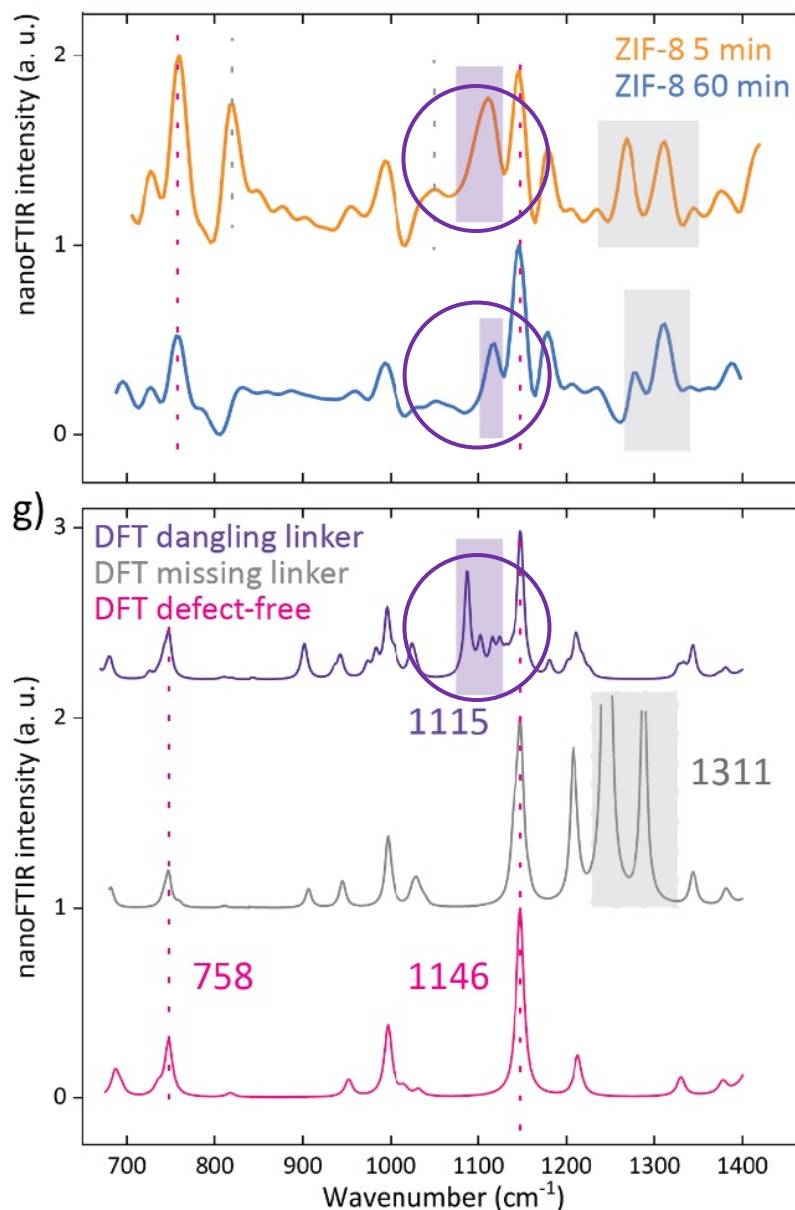
How to model defects in ZIF-8?

- Two point defects simulated at PBEsol0-3c level of theory
- Missing metal Zn-N bonds replaced by N-H bonds
- Missing linker two undercoordinated metal centres are saturated by a water molecule and its conjugated base

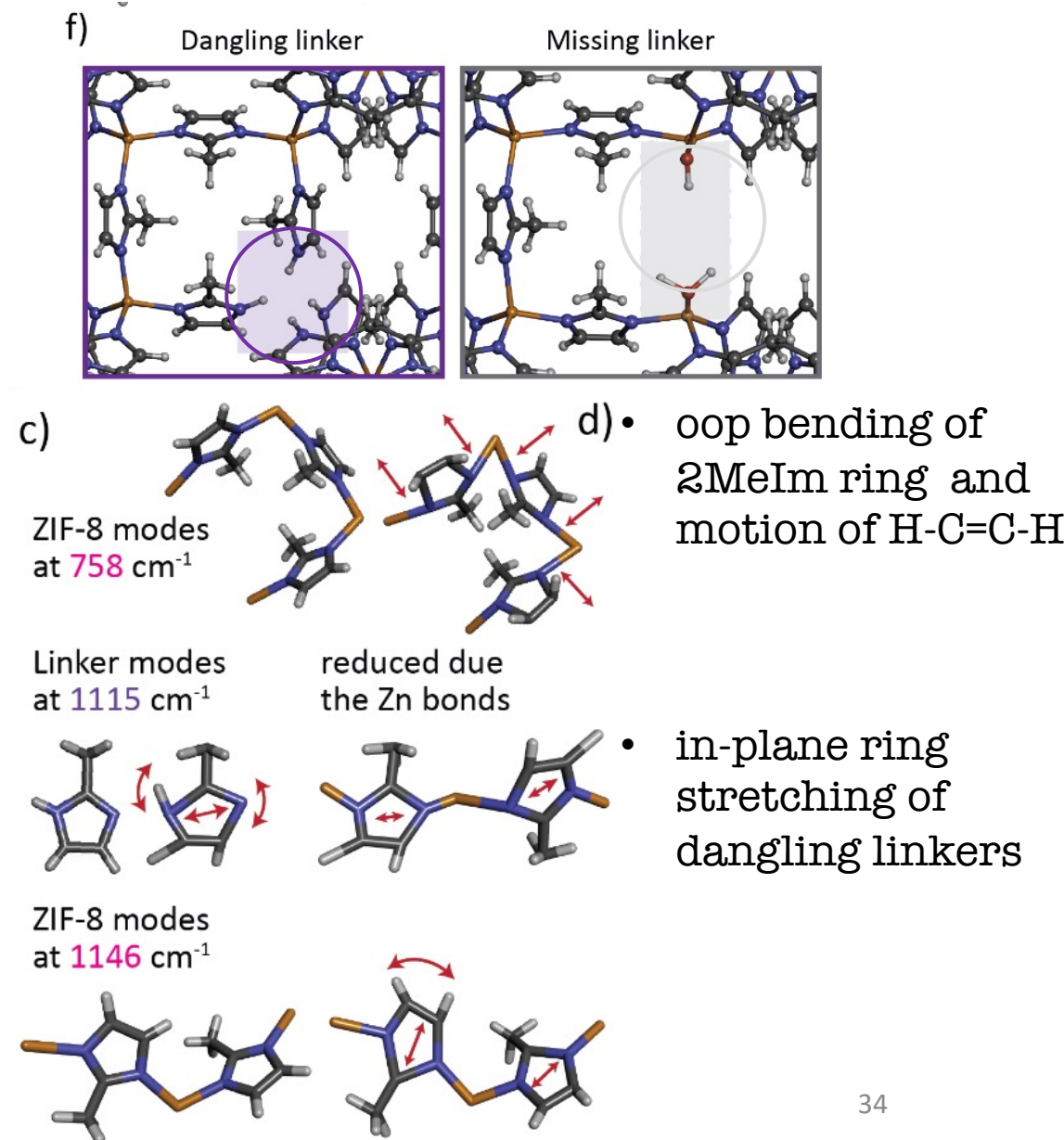
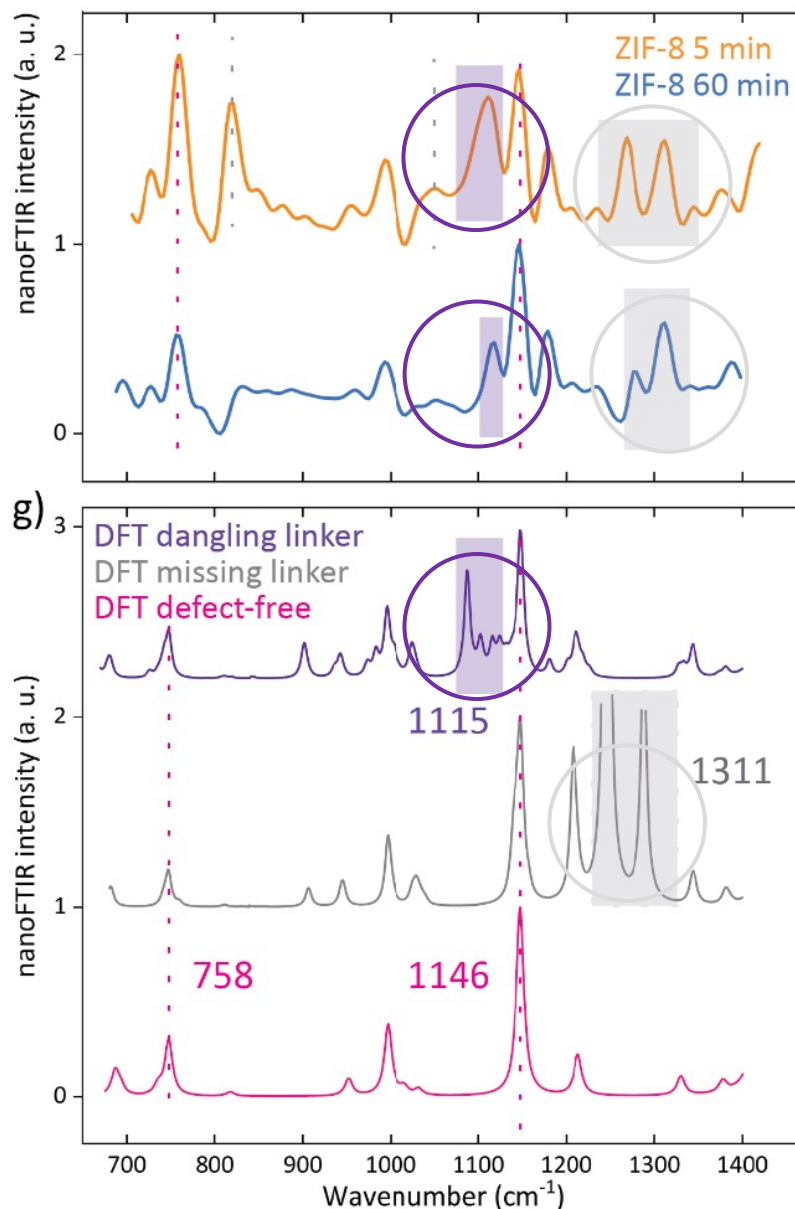
- Comparison between nanoFTIR spectra for 5 and 60 min. growth time and PBEsol0-3c results



- Comparison between nanoFTIR spectra for 5 and 60 min. growth time and PBEsol0-3c results



- Comparison between nanoFTIR spectra for 5 and 60 min. growth time and PBEsol0-3c results



- $C_{ij} > 0$ structures are mechanically stable.
- Defective structures show reduced stiffness and increased anisotropy
- PBEsol0-3c results (**2.70/2.11** GPa) for Young's modulus for defective structures are consistent with TFM data
- **$E(3 \text{ min}) = 1.7 \pm 0.7 \text{ GPa}$**
- **$E(60 \text{ min}) = 2.1 \pm 0.5 \text{ GPa}$**

Elastic Prop.	Defect-free	missing Zn	missing linker
BM (GPa)	10.52	5.60	7.01
E (GPa)	3.33	2.70	2.11
$E_{\text{max/min}}$ (GPa)	3.49 (2.88)	3.48 (1.77)	3.97 (1.94)
$A_E = E_{\text{max}}/E_{\text{min}}$	1.21	2.00	2.05

